

Battlefield Radio Intelligibility Engine

Real-Time On-Device Speech Enhancement Deployed on Snapdragon NPU via Qualcomm AI Hub

Executive Summary

The Battlefield Radio Intelligibility Engine is an ultra-low-latency, stateful streaming neural speech enhancement system engineered specifically for tactical military communications deployed on the Snapdragon X Elite Hexagon NPU. Tactical radio channels suffer from severe acoustic degradation: high-SPL weapon discharges, armored vehicle diesel rumble, rotorcraft turbulence, 300–3400 Hz RF bandpass limitations, Codec2/MELP narrowband quantization, and non-linear push-to-talk (PTT) preamp saturation.

While heavy offline models (e.g., DeepFilterNet3, Demucs) achieve high perceptual scores on synthetic datasets, they cannot execute on edge NPUs due to complex-valued STFT kernels, multi-second latency, and prohibitive power envelopes. Conversely, standard lightweight models frequently suffer from catastrophic speech collapse and musical artifacts at low input SNRs (< 2 dB).

Our engine solves this fundamental trade-off through a 3-stage co-designed architecture:

1. Causal Spatial-Spectral Conv-GRU Neural Core (Branch A): 2-layer stateful recurrent denoiser (6.28 MB FP32 / 6.21 MB INT8) compiled with 100% on-NPU operator residency (240/240 ops) on Snapdragon Hexagon NPU, executing in 411.0 μ s per 32 ms chunk (77.86 $\times$ real-time speedup).
2. Deterministic Soft-Blend Reliability Guardrail: Real-time acoustic safety envelope tracking energy ratio, spectral flatness, and temporal envelope cross-correlation. When low-SNR noise threatens speech formants, it dynamically modulates blending to prevent speech collapse, enabling an Intelligibility Mode that reduces mean WER from 0.443 down to 0.409 (median: 0.353).
3. Host-Side DSP Spectral Post-Filter: Causal Wiener-based residual noise suppressor applied after guard synthesis, eliminating residual radio hiss and musical noise floor flutter without altering neural network weights, enabling a Quality Mode that boosts PESQ to 2.398 (+0.52).

Key Technical Achievements

Dimension	Specification / Result	Verification / Evidence
NPU Execution Efficiency	100.0% on NPU (240/240 operators)	Qualcomm AI Hub Compile Job `jg9zozmlp`
Physical Hardware Latency	411.0 μ s (0.411 ms) median per 32 ms chunk	Qualcomm AI Hub Profile Job `jp1nonj2g`
Real-Time Factor (RTF)	0.01284 (77.86 $\times$ real-time throughput)	Snapdragon X Elite physical hardware
NPU vs CPU Speedup	2.77 $\times$ lower latency vs Host CPU (1.140 ms)	AI Hub Profiler on Snapdragon X Elite
Peak Memory Footprint	13.77 MB runtime memory allocation	Qualcomm AI Hub memory profiler
Stateful Streaming Latency	32 ms algorithmic window (8 ms hop, 4 frames)	Fully causal CausalConv2d + stateful GRU
Quantization Support	INT8 Dynamic Quantization supported	`artifacts/branch_a_denoiser_int8.onnx`
Acoustic Safety Guardrail	Zero Speech Collapse / 0.409 WER	Soft-blend reliability guardrail
Perceptual Quality (Test Set)	PESQ: 2.398 (+0.519 delta) / STOI: 0.944	`results/optimized_pipeline_benchmark.json`

Dual Operational Modes: Quality Mode vs. Intelligibility Mode

Tactical radio deployments demand flexibility between human comprehension and automated ASR systems:

- Intelligibility Mode (`Branch A + Guardrail`): Tailored for automated edge speech recognition (e.g., Whisper) and command-and-control ingestion. Achieves the lowest Word Error Rate (0.409 mean / 0.353 median WER vs. 0.443 / 0.364 baseline) by preserving unvoiced consonants and delicate formant transitions.

- Quality Mode (‘Branch A + Guardrail + Post-Filter’): Tailored for tactical radio operators in noisy combat cockpits or armored vehicles. Delivers a dramatic jump in speech clarity (PESQ 1.879 → 2.398, +0.519 delta; STOI: 0.944) by aggressively squelching background radio hiss, with an operator-selectable trade-off in machine ASR (0.507 WER). Toggleable in real time via Gradio UI or CLI.

Architecture & Signal Flow

[Tactical RF / Mic Input: 16 kHz Audio Stream]



[Causal STFT Analysis Window]

(N_FFT=512, Hop=128, Chunk=4 frames / 32 ms)



Spectral Magnitude Chunk [1, 1, 257, 4]



■ QUALCOMM SNAPDRAGON X ELITE HEXAGON NPU EXECUTION GRAPH ■

■ (240/240 Ops On-NPU | 411 μs Median Latency | QNN Binary)■



■ 1. Causal 2D Spatial-Spectral Convolution Block ■

■ 2. 2-Layer Causal GRU Bottleneck (Hidden Dim = 64) ■

■ - Stateful Hidden State: $h_{in} [2, 1, 64] \rightarrow h_{out}$ ■

■ 3. Sigmoid Spectral Mask Estimator [1, 1, 257, 4] ■

■ 4. Multiplicative Spectral Suppression ■



[Causal iSTFT Synthesis + OLA]



[Deterministic Reliability Guardrail]

(Monitors Energy Ratio, Spectral Flatness, Envelope Correlation)



[Host-Side DSP Spectral Post-Filter]

(Causal Wiener Attenuation & Temporal Smoothing)



Physical Hardware Profiling Data (Qualcomm AI Hub)

All hardware metrics are verified on physical silicon in the Qualcomm AI Hub cloud device farm:

- Target Device: Snapdragon X Elite (Compute Platform)
 - Toolchain: Qualcomm Neural Network (QNN) v2.31 / Qualcomm Hexagon Tensor Processor (HTP)
 - Compile Job ID: `jg9zozmlp` (Status: `Success`, 240/240 ops on NPU, 0 fallback)
 - Profile Job ID: `jp1nonj2g` (Status: `Success`, 240/240 ops on NPU)
 - Profile Summary:
 - Compute Unit: NPU (100%)
 - Median Inference Latency: 411.0 μ s
 - Minimum Inference Latency: 377.0 μ s
 - Maximum Inference Latency: 1421.0 μ s (95th percentile: 496.8 μ s)
 - Processing Budget: 32,000 μ s chunk (Consumes 1.28% of real-time budget, 77.86 $\times$ faster than real-time)
 - Peak Memory: 13.77 MB
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Out-of-Distribution Generalization & Operational Integrity

To validate performance under operational conditions beyond synthetic i.i.d. splits, the engine was benchmarked on extreme out-of-distribution (OOD) scenarios:

1. Armored Vehicle Heavy Rumble (-5.0 dB SNR): Low-frequency tracks and diesel vibration. Guard preserves vowel formants while post-filter cleans rumble.
 2. Urban Breach Firefight (0.0 dB SNR): High-crest gunfire impulse saturation and non-linear PTT clipping.
 3. Rotary-Wing Airlift Turbulence (+3.0 dB SNR): Cyclic rotor blade wash and cockpit acoustic interference.
 4. Narrowband Radio Quantization (+8.0 dB SNR): Codec2 2400 bps vocoder packet loss and RF static squelch tail.
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Conclusion & Impact

The Battlefield Radio Intelligibility Engine proves that state-of-the-art speech enhancement on edge tactical devices does not require massive transformer backbones or high power budgets. By combining:

- A lean, fully NPU-accelerated neural core (411 μ s on Hexagon NPU),
- A mathematical safety guardrail preventing speech collapse and minimizing WER, and
- An operator-toggable DSP spectral post-filter boosting PESQ by +0.52,

the system delivers uncompromising tactical intelligibility, selectable mission modes, and 77 $\times$ real-time throughput on Qualcomm Snapdragon hardware.